

Antoine-Laurent de Lavoisier's meticulous experiments sparked a revolution in science, and his findings had a vital impact on chemistry. His work on combustion and respiration allowed him to determine his law of conservation of mass, and to discover that air was a mixture of gases, including one that he called oxygen.

Born in Paris, Antoine-Laurent de Lavoisier inherited a large fortune at age 5, when his mother died. He qualified as a lawyer but was more interested in science, which he pursued whenever he was able. In 1768, he bought a share in the Ferme Générale, which collected taxes on behalf of the French government, and in 1771, he married the co-owner's 13-year-old daughter, Marie-Anne. She became an invaluable partner in his scientific work, translating chemistry papers into French and assisting him with his experiments.

### Investigating combustion

When he became commissioner of the Royal Gunpowder Administration in 1775, Lavoisier gained access to a laboratory in which he could conduct experiments on combustion. At that time, an incorrect idea called phlogiston theory held that combustible matter such as wood contained "phlogiston," an invisible substance that was released on burning, which explained why substances apparently lost mass when burned. However, Lavoisier found that when he burned sulfur and phosphorus, they acquired mass, suggesting that they were combining with air, and the results could not be explained in terms of phlogiston.

A year earlier, in 1774, Lavoisier had learned from English natural philosopher Joseph Priestley how he had heated mercury calx (mercury oxide) and collected what he called "pure" or "dephlogisticated" air, which burned much more

### MILESTONES

#### ELEMENT FORMS

Proves elements can have different forms by burning a diamond to produce carbon gas in 1772.

#### INVESTIGATES AIR

Hears from Joseph Priestley about "dephlogisticated air" in 1774, and replicates his findings.

#### ANALYZES WATER

In 1783, he reveals that water is a compound of two gases and coins the word "hydrogen."

#### CHEMISTRY TEXTBOOK

Publishes *Elementary Treatise on Chemistry* in 1789—it contains the first table of elements.

### In an experiment,

Lavoisier used a giant magnifying glass to focus the Sun's rays on a diamond in a sealed glass jar. The diamond burned away, but the jar's mass stayed the same, proving his theory of the conservation of mass.

